Attention-Enhanced ResNet34 with Transfer Learning for Brain Tumor MRI Classification

Group 7: 111504006, 111504513, 111504514

Advisor: Prof. Chia-Yu Lin

Motivation

- Brain tumors are severe neurological diseases leading to significant morbidity and mortality if not detected early.
- Magnetic Resonance Imaging (MRI) is the standard non-invasive tool for assessment due to its superior soft-tissue contrast.
- Early and accurate identification is required for effective treatment, which motivates the development of automated and reliable computer-aided diagnosis systems.

Problem Description

- The dependency on manual interpretation creates bottlenecks in the diagnostic workflow, potentially delaying critical treatment.
- As medical imaging data grows, the strain on medical experts becomes unsustainable.
- Existing large architectures can be computationally expensive and prone to overfitting when medical data is scarce or imbalanced.
- On the other hand, smaller models, such as ResNet34, risk underfitting due to limited capacity to capture complex pathological features.

Goal

- To build an accurate and efficient model for brain tumor MRI classification, augmenting diagnostic capabilities.
  - aiming for over 95% accuracy and F1-score above 0.95.
- To provide a fast and consistent second-opinion tool that reduces diagnostic workload.
- To enhance ResNet34 using attention mechanisms while keeping the model lightweight and validated through ablation studies.

Related Works

Deep and Transfer Learning

- Deep Learning
  - CNN like VGG, ResNet, and EfficientNet are mainstream approaches for brain tumor classification.
- Transfer Learning
  - Leverages knowledge from large-scale source domains to target domains, reducing training time and resource needs.
  - Fine-tunes all layers, achieving faster stabilization and recognizing task-specific patterns.

Data Augmentation

- Essential for preventing overfitting and improving generalization in medical imaging where data is scarce or imbalanced.
- Includes classic geometric transformations, such as flipping and resized crop.

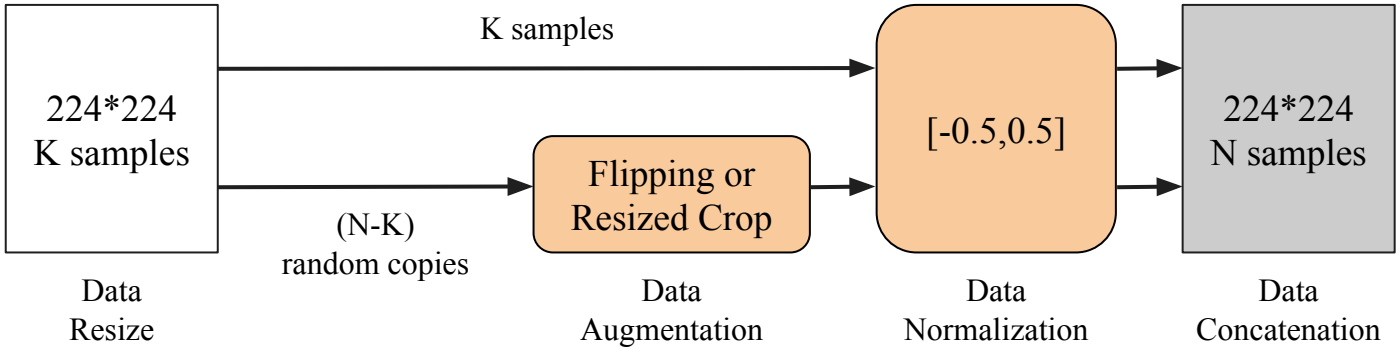
Attention Mechanisms

- SE-Net (Squeeze-and-Excitation) introduces channel attention, helping the model focus on important feature channels.
- CBAM (Convolutional Block Attention Module) combines channel and spatial attention, improving feature selection in both dimensions.
- These modules are widely used to boost CNN performance by suppressing irrelevant information.

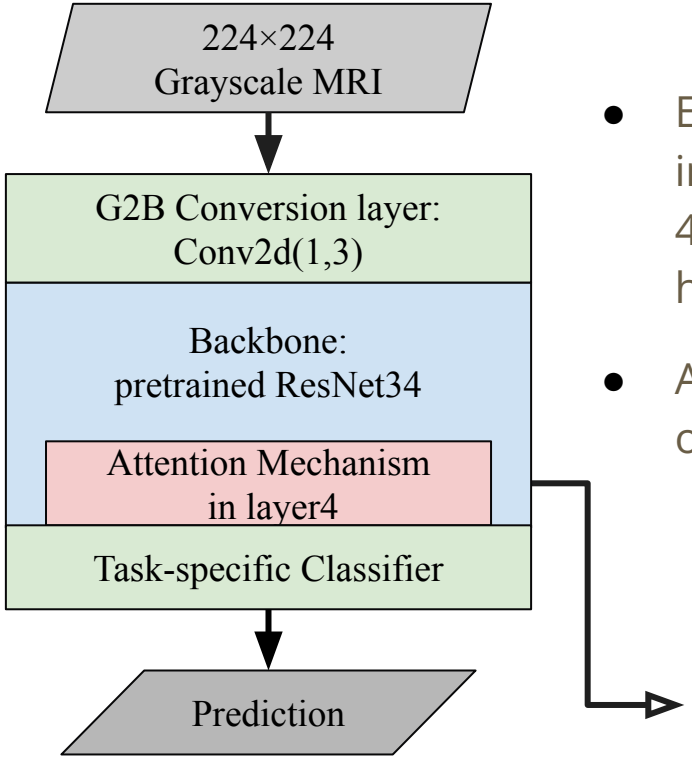
Proposed Method

Data Preprocessing

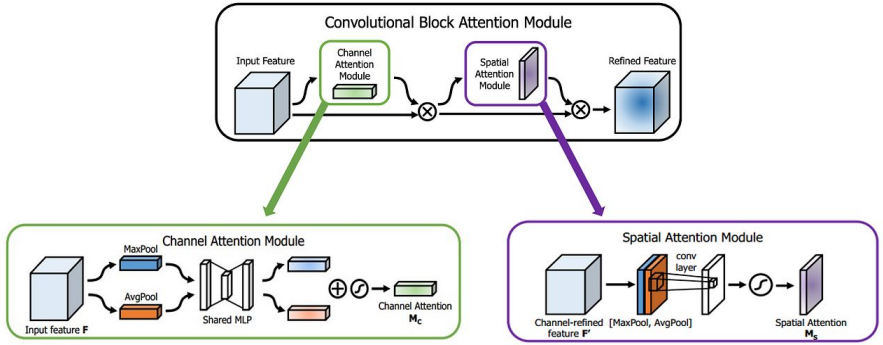
- Original dataset is imbalanced across tumor types.
- Use augmentation (horizontal/vertical flip, random resized crop) to generate missing images.
  - until 1600 training samples and 400 test samples for each class.
- All images are resized to 224*224 and normalized.



Model Architecture



- Enhance ResNet34 by adding gray-to-RGB conversion at input, attention mechanism in the last BasicBlock of the layer 4, and replacing the original classifier with a task-specific head.
- Attention mechanism can be channel-based (SE), spatial, or both (CBAM).



Experiments

Dataset

- Brain Tumor MRI Dataset released by Masoud Nickparvar.
  - A composite dataset of 7,023 brain MRI images was used, combining Figshare, SARTAJ, and Br35H datasets.
  - Images (training+test) are classified into 4 classes:
    glioma (1321+300), meningioma (1339+306), pituitary (1457+300), and no tumor (1595+405).
- After augmenting the data to (1600+400) samples for each class, we employed 5-fold cross-validation with an 80:20 train-validation split.

Configuration & Evaluation Metrics

- Environment: Experiments were conducted using Google Colab.
- Hyperparameters
  - Learning rate: 0.00005
  - Batch size: 8
  - Epochs: 5
- Evaluation Metrics
  - Accuracy = (TP+TN)/(TP+TN+FP+FN)
  - Precision = TP/(TP+FP)
  - Recall = TP/(TP+FN)
  - F1-score = 2*Precision*Recall/(Precision+Recall)

Results

- The proposed TResNet34_CBAM achieves 98.06% accuracy, outperforming the baseline ResNet34 by 6.68 percentage points.
  - All metrics-precision, recall, and F1-score-consistently reach 0.98, demonstrating balanced performance.
- The baseline ResNet34 establishes solid performance at 91.38%. However, EfficientNet models, despite being parameter-efficient, achieve only ~84% accuracy, suggesting they may underfit the complex tumor patterns.

	Accuracy(%)	Precision	Recall	F1-scroe
(Baseline) ResNet34	91.38	0.91	0.91	0.91
EfficientNet_B0	84.81	0.85	0.85	0.85
EfficientNet_B1	84.00	0.85	0.84	0.84
(Ours) TResNet34_CBAM	**98.06**	**0.98**	**0.98**	**0.98**

Results

- The proposed TResNet34_CBAM outperforms baseline models for two key reasons:
  - Transfer learning accelerates convergence and improves generalization ability, making the network learn faster, more accurately, and less prone to overfitting.
  - The CBAM attention mechanism helps the network emphasize the most informative tumor regions while suppressing irrelevant background, which standard CNNs often fail to capture adequately.

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Results - Ablation Study

- Starting from the baseline TResNet34 without attention at 96.92% accuracy, we observe progressive improvements: channel-based attention adds 0.39%, spatial attention contributes 0.96%, and the combined CBAM approach achieves the highest 98.06% accuracy with 0.98 across all metrics.

	Attention	Accuracy(%)	Precision	Recall	F1-scroe
TResNet34_None	None	96.92	0.97	0.97	0.97
TResNet34_ChAt	Channel-based	97.31	0.97	0.97	0.97
TResNet34_SpAt	Spatal	97.88	0.98	0.97	0.97
TResNet34_CBAM	Channel+Spatial	**98.06**	**0.98**	**0.98**	**0.98**

Results - Ablation Study

- The channel attention improves feature selection and the spatial attention improves focus on tumor location. CBAM, which combines the two, thus achieves the effect of reducing background interference and more accurately pinpointing tumor location.

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Conclusion

- We introduced an Attention-Enhanced ResNet34 framework for brain tumor MRI classification, combining:
  - Balanced Data Augmentation to stabilize training across classes.
  - Transfer Learning for efficient feature reuse.
  - Channel & Spatial Attention (CBAM) for better feature refinement.
- Our proposed mode:
  - reached 98.06% Accuracy and 0.98 for all metrics, outperforming baseline ResNet34, EfficientNets, and ablation versions.
  - Demonstrated that adding CBAM attention provides the most consistent improvement in classification performance.

Conclusion

- Our work provides:
  - more accurate and reliable tumor classification, potentially assisting clinicians in early diagnosis.
  - lightweight and efficient architecture, making it suitable for real-world deployment, even on limited hardware.
  - reduced development cost and training time through transfer learning and balanced augmentation.